Spring 2026

CS 3503 Assignment – 6

Memory & ISA MARIE

1. Consider the MARIE program below. (10 Points)
   a) List the hexadecimal code for each instruction.

Address Instruction Opcode Hex Address (hex)

100	LOAD X	1	105
101	ADD Y	3	106
102	STORE Z	2	107
103	HALT	7	000
104	X, DEC 10		
105	Y, DEC 15		
106	Z DEC 0		

b) Draw the symbol table.

X	104
Y	105
Z	106

c) What is the value stored in the AC when the program terminates?

25. It holds the value of 10 before ending and then it adds the value of 15.

Hex Address	Label	Instruction
100	Start,	LOAD A
101		ADD B
102		STORE D
103		CLEAR
104		OUTPUT
105		ADDI D
106		STORE B
107		HALT
108	A,	HEX 00FC
109	B,	DEC 14
10A	C,	HEX 0108
10B	D,	HEX 0000

2. Write the following code segment in MARIE's assembly language: (20 Points)
   a) if X > 1 the

      Y = X + X;

      X = 0;

      endif;

      Y = Y + 1;

LOAD X

SUBT ONE / Subtract 1 to check if X > 1

SKIPCOND 800 / If AC > 0 (meaning X > 1), skip the jump to endif

JUMP ENDIF / If condition was false, go to the end

/ IF BLOCK

LOAD X

ADD X

```
STORE Y
CLEAR
STORE X

ENDIF, LOAD Y
ADD ONE
STORE Y
HALT

ONE, DEC 1
X, DEC 5    / Example value
Y, DEC 0
```

b) X = 1;

while X < 10 do

X = X + 1;

Endwhile

```
/ Initialize
LOAD ONE
STORE X

WHILE, LOAD X
SUBT TEN      / Check X - 10
SKIPCOND 000   / If AC < 0 (X < 10), keep going (skip the jump to exit)
```

JUMP ENDWHILE / Exit loop if X >= 10

/ LOOP BODY

LOAD X

ADD ONE

STORE X

JUMP WHILE   / Repeat

ENDWHILE, HALT

ONE, DEC 1

TEN, DEC 10

X, DEC 0

3. Write a MARIE program using a loop that multiplies two positive numbers by using repeated addition. For example, to multiple 3 x 6, the program would add 3 six times, or 3+3+3+3+3+3.
(15 Points)

LOAD ZERO

STORE RESULT

WHILE, LOAD MULTIPLIER

SKIPCOND 800

JUMP ENDWHILE

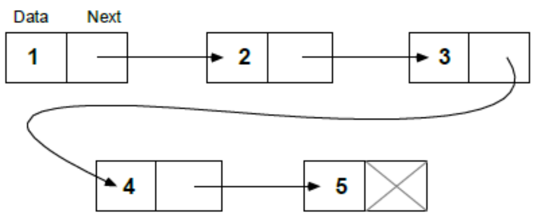
```
LOAD RESULT
ADD MULTIPLICAND
STORE RESULT

LOAD MULTIPLIER
SUBT ONE
STORE MULTIPLIER
JUMP WHILE

ENDWHILE, LOAD RESULT
HALT

MULTIPLICAND, DEC 3
MULTIPLIER,   DEC 6
RESULT,       DEC 0
ZERO,         DEC 0
ONE,          DEC 1
```

4. A linked list is a linear data structure consisting of a set of nodes, where each one except the last one points to the next node in the list. (Appendix A provides more information about linked lists.) Suppose we have the set of 5 nodes shown in the illustration below. These nodes have been scrambled up and placed in a MARIE program as shown below. Write a MARIE program to traverse the list and print the data in order as stored in each node (15 Points)



MARIE program fragment:

```
Address
  (Hex)
  00D |  Addr,   Hex ????       / Top of list pointer:
                                / You fill in the address of Node1.
  00E |  Node2,  Hex 0032       / Node's data is the character "2."
  00F |          Hex ????       / Address of Node3.
  010 |  Node4,  Hex 0034       / Character "4."
  011 |          Hex ????
  012 |  Node1,  Hex 0031       / Character "1"
  013 |          Hex ????
  014 |  Node3,  Hex 0033       / Character "3"
  015 |          Hex ????
  016 |  Node5,  Hex 0035       / Character "5"
  017 |          Hex 0000       / Indicates terminal node.
```

```
LOAD HEAD     / add in the first node

STORE PTR     / Initialize the pointer


LOOP, LOAD PTR

SKIPCOND 800  / If PTR is ), we are at the end stop

JUMP DONE


LOADI PTR     / Load the data from the current node

ADD SUM

STORE SUM


LOAD PTR

ADD ONE       / Move to the next pointer address
```

```
STORE PTR
LOADI PTR     / Load the address of the next node
STORE PTR     / Update PTR with the next node's address
JUMP LOOP

DONE, LOAD SUM
HALT

/ DATA
HEAD,  HEX 200
PTR,   HEX 0
SUM,   DEC 0
ONE,   DEC 1
```

5. More registers appears to be a good thing, in terms of reducing the total number of memory accesses a program might require. Give an arithmetic example to support this statement. First, determine the number of memory accesses necessary using MARIE and the two registers for holding memory data values (AC and MBR). Then perform the same arithmetic computation for a processor that has more than three registers to hold memory data values. (15 Points)

Going based off of example used in class

Expression : Z= A * B + C * D + E* F

```
LOAD R1 A
MULT R1 B
LOAD R2 C
MULT R2 D
LOAD R3 E
```

MULT R3 F

ADD R1, R2

ADD R1, R3

STORE Z R1

6. Show how the following values would be stored by byte-addressable machines with 32-bit words, using little endian and then big endian format. Assume each value starts at address 0x10. Draw a diagram of memory for each, placing the appropriate values in the correct (and labeled) memory locations. (10 Points)

a) 0x456789A1

Type	10	11	12	13
Big Endian	45	67	89	A1
Little Endian	A1	89	67	45

b) 0x0000058A

Type	10	11	12	13
Big Endian	00	00	05	8A
Little Endian	8A	05	00	00

7. Write a MARIE program that performs the three basic stack operations: push, peek, and pop (in that order). In the peek operation, output the value that's on the top of the stack. (If you are unfamiliar with stacks, see Appendix A for more information.) (15 Points)

/ PUSH OPERATION

LOAD VAL   / Load the value we want to put on the stack

```
STOREI SP       / Store the value at the address held in the stack pointer
LOAD SP
ADD ONE         / Move the Stack Pointer to the next empty slot
STORE SP

/ PEEK OPERATION
LOAD SP
SUBT ONE        / Go back to the address of the top item
STORE ADDR      / Store that temporary address
LOADI ADDR      / Load the value from the top of the stack
OUTPUT          / Display the value on the screen

/ POP OPERATION
LOAD SP
SUBT ONE        / Remove the top item
STORE SP
HALT

/ DATA SECTION
VAL,   DEC 15   / The value that needs to be shown
SP,    HEX 200  / Start the stack at address 200
ADDR,  HEX 0    / Temporary address for the Peek operation
ONE,   DEC 1    / Constant for math
```

Submission Requirements

- You may complete the assignment in either of the following ways:

  Write your responses directly in this document, or

  Write your responses by hand, take clear photos, and insert the photos into this same document.

- Only ONE document should be submitted.

  Acceptable formats: PDF or Word document.

- Multiple image files submitted separately (without being combined into a single PDF or document) will NOT be accepted and will receive a grade of zero.

- Ensure that all work is clearly labeled, legible, and organized according to the question numbers.